№6. $I(\lambda)=\int\limits_{-\infty}^{\infty} e^{-\lambda(x^2-3ix)} F(x)\,dx$, $F(x)=\int\limits_{0}^{\infty} \frac{(1+ix)\,y^{ix}}{(1+y)^{2+2ix}}\, e^{-y}\, dy$

$f(x)=x^2-3ix$

$f'(x)=2x-3i=0$

$x_0=\frac{3i}{2}$

$f''(x)=2$, $\arg f=0$

$2\alpha+0=\pi+2\pi n$

$\alpha=\frac{\pi}{2}+\pi n$

$f\left(\frac{3i}{2}\right)=-\frac{9}{4}+\frac{5}{2}=\frac{9}{4}$

$g(y)=\frac{y^{ix}}{(1+y)^{2+2ix}}e^{-y}$, $y\to0$ $g(y)\to y^{ix}e^{-y}$

$F\left(\frac{3i}{2}\right)=\int\limits_0^\infty \left(1-\frac{3}{2}\right)y^{-\frac{3}{2}}e^{-y}\,dy=-\frac{1}{2}\cdot\Gamma\left(-\frac{1}{2}\right)=\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$

$I(\lambda)=F(x_0)\,e^{\lambda f_0}\sqrt{\frac{2\pi}{\lambda f''(x_0)}}$

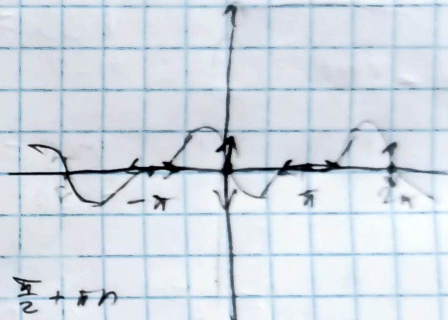
$I(\lambda)=\sqrt{\pi}\cdot\sqrt{\frac{2\cdot\pi}{\lambda\cdot 2}}\cdot e^{\lambda\frac{9}{4}}$

$I(\lambda)=\frac{\pi}{\sqrt{\lambda}}e^{\lambda\frac{9}{4}}$

N7. $I(\lambda) = \int\limits_{-\infty}^{\infty} \cos(\lambda \cos x) \frac{\sin x}{x} dx$ $\qquad \cos x = Re[e^{ix}]$

$\cos(\lambda \cos x) = Re[e^{i\lambda \cos x}]$

$I(\lambda) = Re \int\limits_{-\infty}^{\infty} \frac{\sin x}{x} e^{i\lambda \cos x} dx$



$f(x) = \cos x$

$f'(x) = -\sin x = 0$

$x = \pi n$

$f''(x) = -\cos x$

$f''(x_0)_1 = -1, \quad \arg f'' = \pi, \quad \alpha = \frac{\pi}{2} + \pi n$

$f''(x_0)_2 = 1 \quad \arg f'' = 0, \quad \alpha = \pi n$

$x_1 = 0 \; : \; \cos x \approx 1 - \frac{x^2}{2} \; ; \; \sin x \approx x$

$x_2 = \pi$

$I(\lambda) \approx Re \int\limits_{-\infty}^{\infty} \frac{x}{x} \cdot e^{i\lambda(1 - \frac{x^2}{2})} dx = Re\left(e^{i\lambda} \int\limits_{-\infty}^{\infty} e^{-\frac{i\lambda x^2}{2}} dx\right)$

$\int\limits_{-\infty}^{\infty} e^{\frac{i\lambda x^2}{2}} dx = \sqrt{\frac{2\pi}{i\lambda}} = \sqrt{\frac{2\pi}{\lambda}} e^{-i\frac{\pi}{4}}$

$I(\lambda) \approx Re\left(e^{i(\lambda - \frac{\pi}{4})} \sqrt{\frac{2\pi}{\lambda}}\right) = \sqrt{\frac{2\pi}{\lambda}} \cos(\lambda - \frac{\pi}{4})$